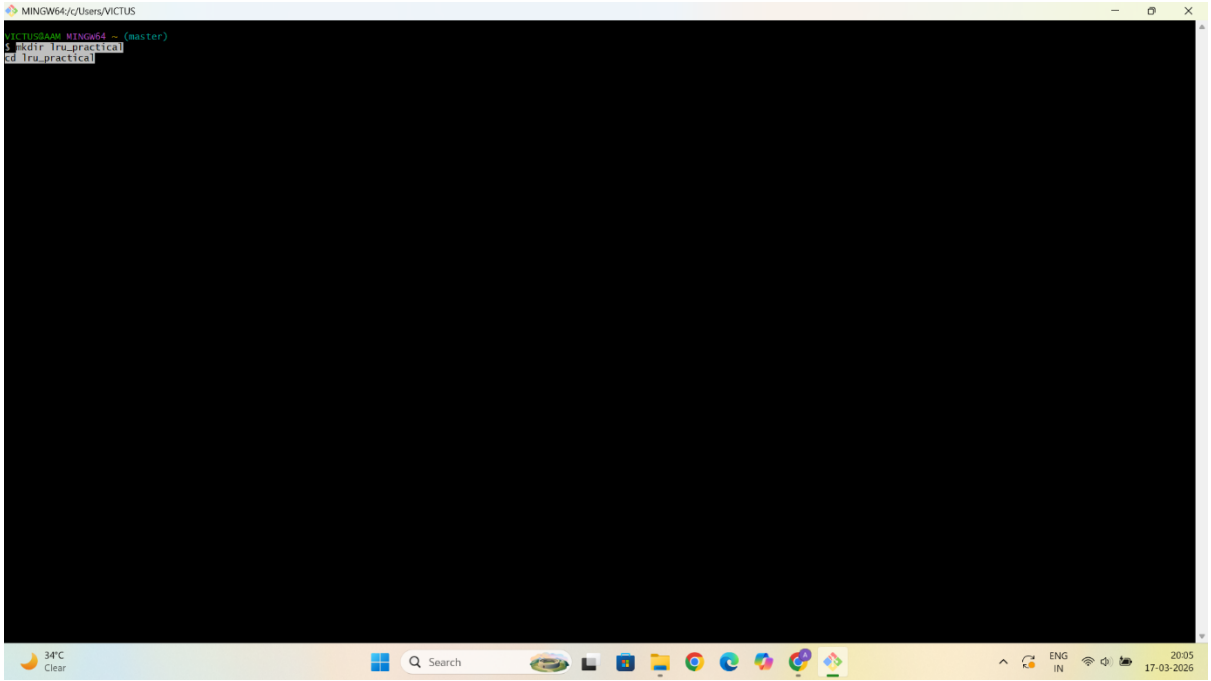
Practical No. 8

Aim

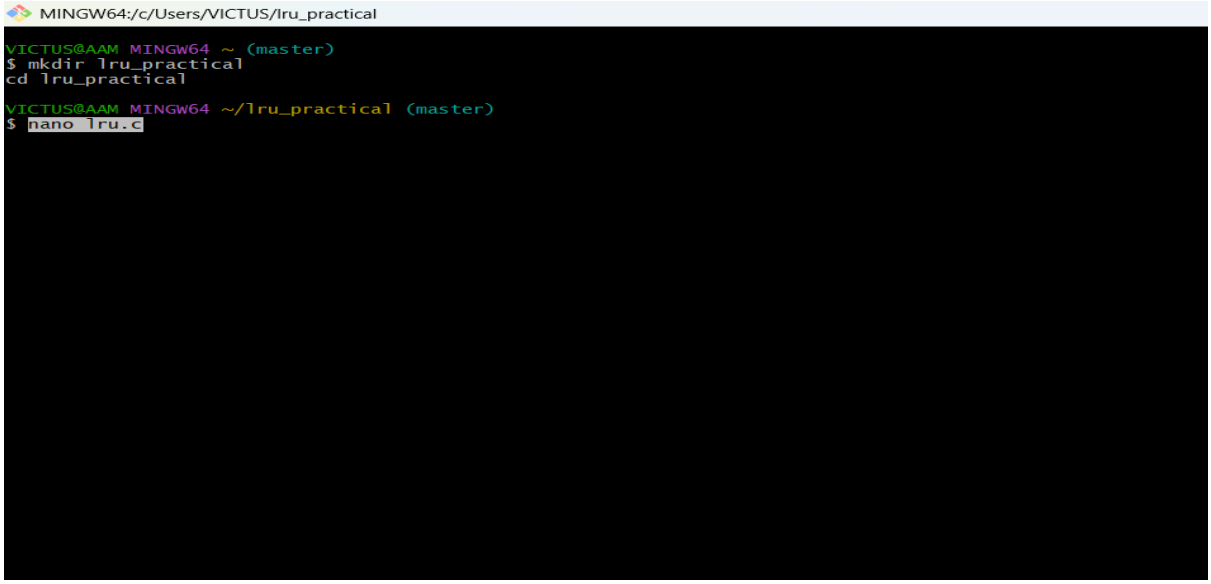
*To write and execute a program to calculate **page hits and page faults** using **LRU (Least Recently Used) page replacement algorithm**.*

***Step 1:** Open Git Bash*
Git Bash was opened to execute commands and run the program.

***Step 2**: Create Directory*
A new directory was created for the practical:



Step 3: *Create C File A C file was created using nano editor*



Step 4: *Write Program*

The LRU Page Replacement program was written in C language with:

- *Number of frames = 3*
- *Page reference string = 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5The program calculates total page hits and page faults.*

MINGW64:/c/Users/VICTUS/lru_practical

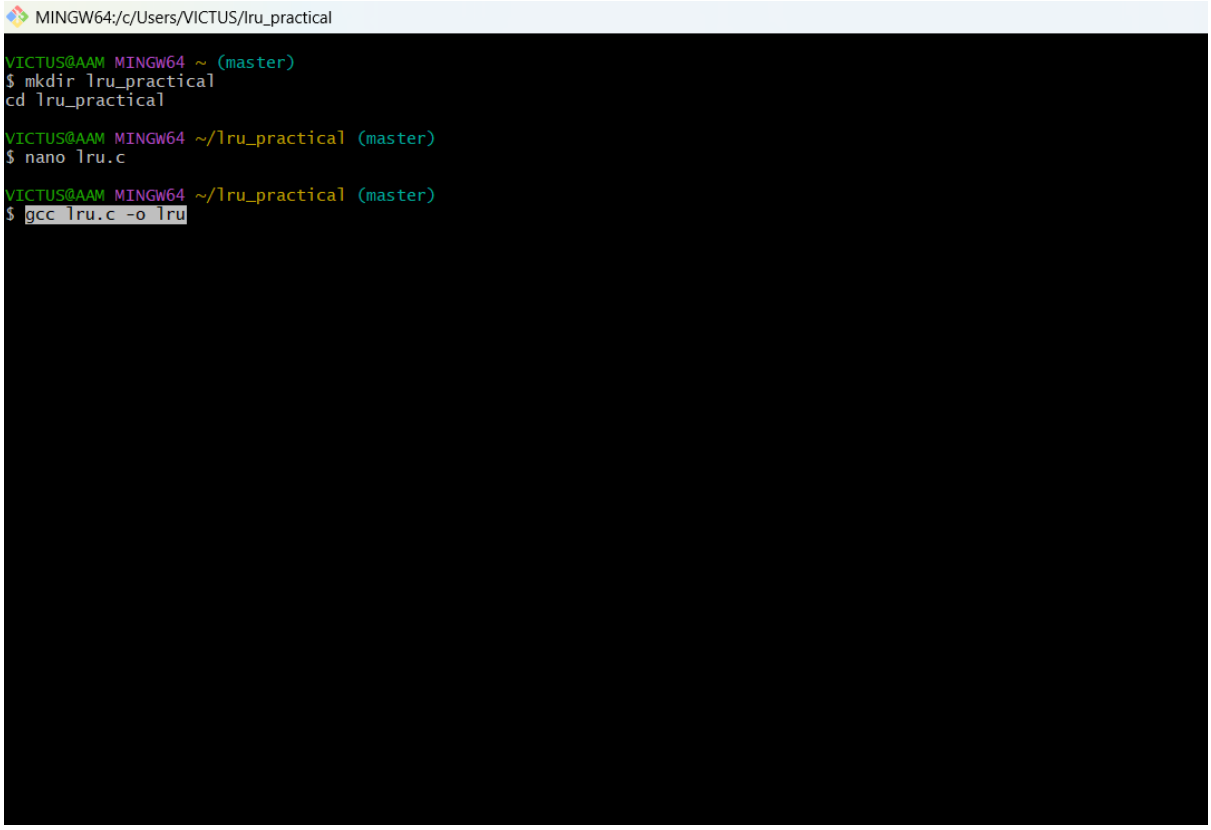
```
  GNU nano 8.7                                                    lru.c
#include <stdio.h>

int main() {
    int frames = 3;
    int pages[] = {1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5};
    int n = 12;

    int memory[3];
    int time[3];
    int i, j;
    int hits = 0, faults = 0;
    int counter = 0;

    // initialize memory
    for(i = 0; i < frames; i++) {
        memory[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int page = pages[i];
        int found = 0;

        // check hit
        for(j = 0; j < frames; j++) {
            if(memory[j] == page) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        // page fault
        if(!found) {
            int lru_index = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < time[lru_index]) {
                    lru_index = j;
                }
            }

            memory[lru_index] = page;
            counter++;
            time[lru_index] = counter;
            faults++;
        }
    }

    printf("Total Hits = %d\n", hits);
    printf("Total Faults = %d\n", faults);

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo       M-A Set Mark   M-] To Bracket  M-
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo       M-6 Copy       ^B Where Was    M-
```

MINGW64:/c/Users/VICTUS/lru_practical

```
  GNU nano 8.7                                                    lru.c
    int i, j;
    int hits = 0, faults = 0;
    int counter = 0;

    // initialize memory
    for(i = 0; i < frames; i++) {
        memory[i] = -1;
        time[i] = 0;
    }

    for(i = 0; i < n; i++) {
        int page = pages[i];
        int found = 0;

        // check hit
        for(j = 0; j < frames; j++) {
            if(memory[j] == page) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        // page fault
        if(!found) {
            int lru_index = 0;

            for(j = 1; j < frames; j++) {
                if(time[j] < time[lru_index]) {
                    lru_index = j;
                }
            }

            memory[lru_index] = page;
            counter++;
            time[lru_index] = counter;
            faults++;
        }
    }

    printf("Total Hits = %d\n", hits);
    printf("Total Faults = %d\n", faults);

    float hit_ratio = (float)hits / n * 100;
    float fault_ratio = (float)faults / n * 100;

    printf("Hit Percentage = %.2f%%\n", hit_ratio);
    printf("Fault Percentage = %.2f%%\n", fault_ratio);

    return 0;
}

^G Help        ^O Write Out   ^F Where Is    ^K Cut         ^T Execute     ^C Location    M-U Undo
^X Exit        ^R Read File   ^\ Replace     ^U Paste       ^J Justify     ^/ Go To Line  M-E Redo
```

Step 5: *Compile Program*

The program was compiled using GCC compiler:



Step 6: Output Observed

The output obtained is:

- *Total Hits = 2*
- *Total Faults = 10*
- *Hit Percentage = 16.67%*
- *Fault Percentage = 83.33%*

MINGW64:/c/Users/VICTUS/lru_practical

```
VICTUS@AAM MINGW64 ~ (master)
$ mkdir lru_practical
cd lru_practical

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ nano lru.c

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ gcc lru.c -o lru

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ ./lru
Total Hits = 2
Total Faults = 10
Hit Percentage = 16.67%
Fault Percentage = 83.33%

VICTUS@AAM MINGW64 ~/lru_practical (master)
$ |
```

The program for LRU page replacement algorithm was successfully executed, and the page hits and page faults were calculated correctly.